

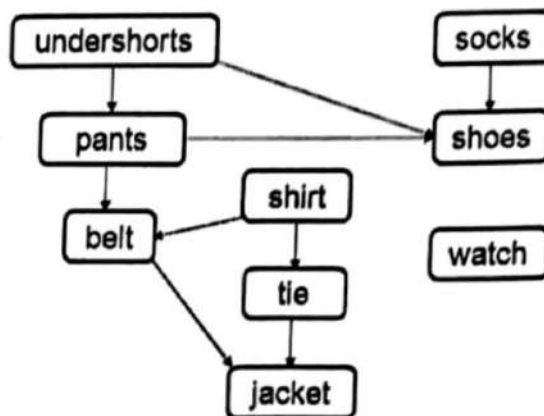
Time: 1 hour 20 minutes

Note: Answer all questions. Figure in the right margin indicate marks.

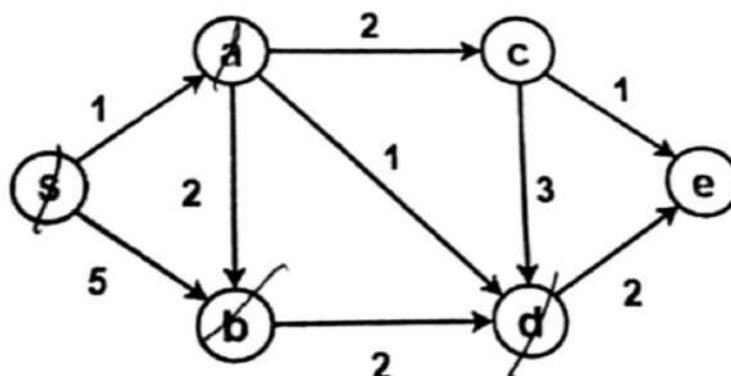
1. Draw a binary search tree using the following numbers. Also, show the linked representation of the BST. 6
~~37~~, ~~2~~, 45, ~~48~~, ~~41~~, ~~29~~, 20, 30, ~~49~~, 70
2. What is complete binary tree? Find the Huffman tree of the following 7
 information.

Letter	Freq
a	120
d	42
l	42
u	37
c	32
m	24
k	7
z	2

3. What is the minimum spanning tree? Find the discovery and finishing 7
 time of every node in following graph. Also find the forward and cross edge:



4. Explain the condition under which the Bellman-Ford algorithm cannot 10
 determine shortest paths. Find the single source shortest path using
 Dijkstra Algorithm. Assume source is S.



East West University

Department of Computer Science and Engineering

Course Code: CSE 207

Title: Data Structures

Course Instructor:

Manowarul Islam

Date: 12.11.2025

Fall 2025

Mid-Term Examination

Section: 10

Marks:30

Time: 1 hour 20 minutes

Note: Answer all questions. Figure in the right margin indicate marks.

1. Write a simple procedure to search an element into a sorted linked list. CO1-4
2. Mention the application of stack. Convert the following expression to equivalent postfix expression using stack: $A+B*(C^D-E)^{(F+G*H)}-I$ CO2-10
3. Compare how linked list can be more useful for item insertion and deletion operation than an array. 3/CO1-6
4. What are the properties of recursive function? What are the properties of recursive function? State the solution of tower of Hanoi problem for $N=4$. 4/CO2-5
5. Suppose you have a circular queue of size 6 (1 to 6). Initially the value of the front and rear is 0. The current state of the queue is shown in the figure where the front=4 and the rear=6. Now show the state of the queue if the following operation is performed on the queue. Also find the value of front and rear in each step. 5/CO1-5

1	2	3	4	5	6
			12	72	35
			Front=4		Rear=6

insert (10); insert (5), delete(), delete(), insert (9); insert (9); insert (17); insert (15); insert (18); insert (11); insert (18).

$\frac{A \overline{B} C B A A C}{A \overline{B} C / C \overline{A} / A / A}$

East West University

Department of Computer Science and Engineering

Course Code: CSE 207 Title: Data Structures

Course Instructor: Manowarul Islam

Date: 30.07.2025 Summer 2025

Mid-Term Examination Section: 11

Marks:30

Time: 1 hour 20 minutes

Note: Answer all questions. Figure in the right margin indicate marks.

1 Write a simple procedure to insert an element into a circular queue. CO1-4

X What is stack? Convert the following expression to equivalent postfix expression using stack: $K + L - M * N + (O \wedge P) * W / U / V * T + Q$ CO2-8

X You are given the following **sorted array** of integers: $A = [5, 9, 12, 17, 23, 28, 34, 39, 45, 51, 60, 72]$. What is the index of 70 when using the ternary search algorithm? CO1-6 4

X How linked list is more efficient for insertion and deletion operation compare to array. CO2-5

OR

What are the properties of recursive function? Write the solution for Tower of Hanoi problem for $N=4$. CO2-5

X What is the problem of linear queue? What is queue? Suppose you have a circular queue of size 6 (1 to 6). Initially the value of the front and rear is 0. Now show the state of the queue if the following operation is performed on the queue. Also find the value of front and rear in each step. CO1-7

~~insert (10); insert (5), delete(), delete(), insert (9); insert (9); insert (7);~~
~~insert (5); insert (8); insert (10); insert (11).~~

East West University
Department of Computer Science and Engineering
Course Code: CSE 207 Title: Data Structures
Course Instructor: Manowarul Islam
Date: 30.07.2025 Summer 2025
Mid-Term Examination Section: 10
Marks:30

Time: 1 hour 20 minutes

Note: Answer all questions. Figure in the right margin indicate marks.

- 1 Write a simple procedure to delete an element into a circular queue. CO1-4
- 2 Write an algorithm to evaluate a postfix expression. Evaluate the following postfix expression using your algorithm. CO2-7
3, 5, +, 6, 4, -, *, 4, 1, -, 2, ^, +
- 3 What are the properties of binary search algorithm? You are given the following sorted array of integers: A = [5, 9, 12, 17, 23, 28, 34, 39, 45, 51, 60, 72]. What is the index of 72 when using the ternary search algorithm? CO1-7
- 4 What are the advantages of using linked list over an array? Justify your answer. CO2-5

OR

What are the properties of recursive function? Write a recursive function that calculates the sum of all integers from 1 to N. CO2-5

- The function should take a single integer N as input.
- It should return the sum of all numbers from 1 up to N.
- If N is 0, the function should return 0.

- 5 What is the problem of linear queue? Suppose you have a circular queue of size 7 (1 to 7). Initially the value of the front and rear is 0. Now show the state of the queue if the following operation is performed on the queue. Also find the value of front and rear in each step. CO1-7

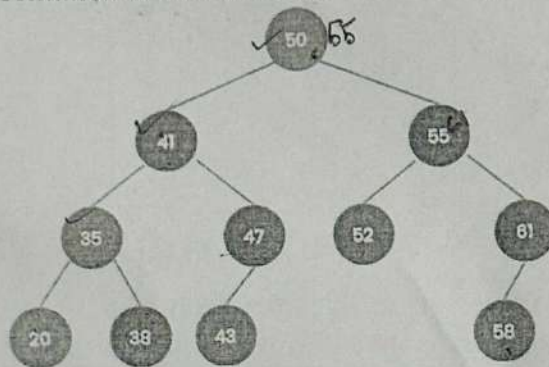
delete(), delete(); insert(10); insert(15); insert(9); insert(7); insert(17); delete(); insert(8); delete(); insert(11); insert(13); insert(11).

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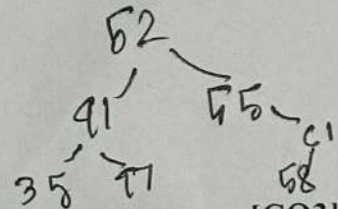
Department of Computer Science and Engineering
 Course Code: CSE 207 Title: Data Structures
 Course Instructor: Manowarul Islam
 Date: 26.05.2025 Spring 2025
 Final Examination Section: 9
 Time: 1 hour 20 minutes Marks: 20

Note: Answer all questions. Figure in the right margin indicate marks.

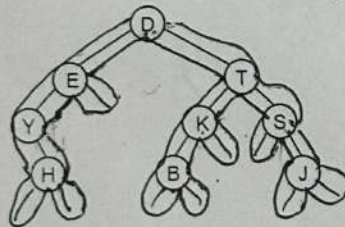
1. State the properties of recursion. Delete 50 from the following tree. Show [CO2]-
 the link representation before and after the delete operation. 6



55 node left right



2. Define complete binary tree. Consider the following tree.
 i. Shows the sequential memory representation of the tree.
 ii. Show the in-order and post-order traversal of the tree.



or

In: H, Y, E, D, B, K, T, S, J,
 Post: H, Y, E, B, K, J, S, T, D

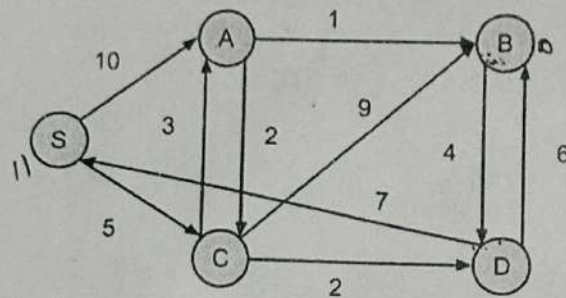
What are the application areas of Huffman coding? Find the code of the following info using Huffman algorithm.

Symbol	Frequency
A	11
B	12
C	13
D	14
E	24
F	26

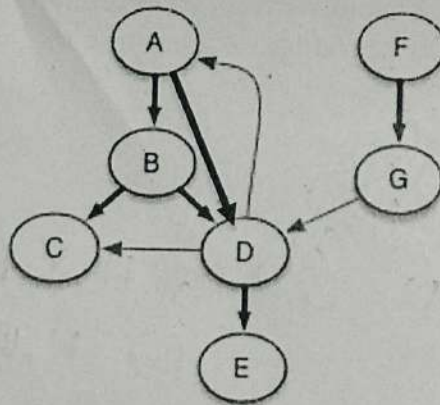
H, Y, E, B, K, J, S, T, D

B	A	B	E	D	S
D	∞	0	∞	∞	
S	∞		∞	∞	
E	∞				
A	21				

3. What is a negative weight cycle in graph? Find the single source shortest path using Dijkstra Algorithm. Assume source is B. [CO3]-
8



4. Show the d and f values that result from running the Depth-first search (DFS) on the directed graph given below. Also find the forward or cross edge. [CO2]-
7



Define MST? Distinguish between prim's and kruskal's algorithm.

[CO3]-
5

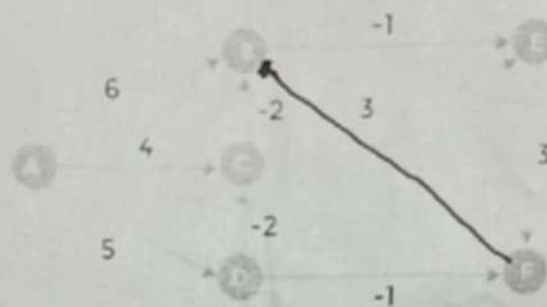
Minimal spanning tree

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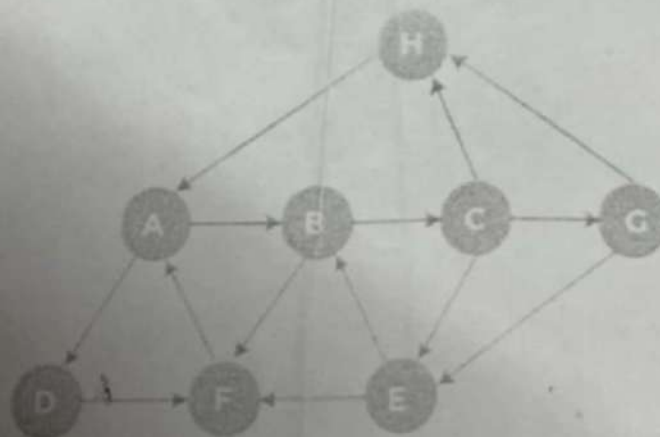
Department of Computer Science and Engineering
Course Code: CSE 207 Title: Data Structures
Course Instructor: Manowarul Islam
Date: 13.09.2023 Summer 2023
Final Examination Section: 6
Time: 1 hour 20 minutes Marks: 20

Note: Answer all questions. Figure in the right margin indicate marks.

1. Describe relaxation process with example. Find all pair shortest path of the following graph using bellman-ford algorithm. [CO2]-6.5



2. Differentiate between Prim's and Kruskal's algorithm. [CO2]-2.5
3. State how bellman-ford can identify negative cycle in a graph. [CO3]-3
4. Show the d and f values that result from running the Depth-first search (DFS) on the directed graph given below. Finally find the back edge, cross-edge and forward edge of the graph. Now from your results, state how DFS can detect any cycle in the graph. [CO3]-7.5

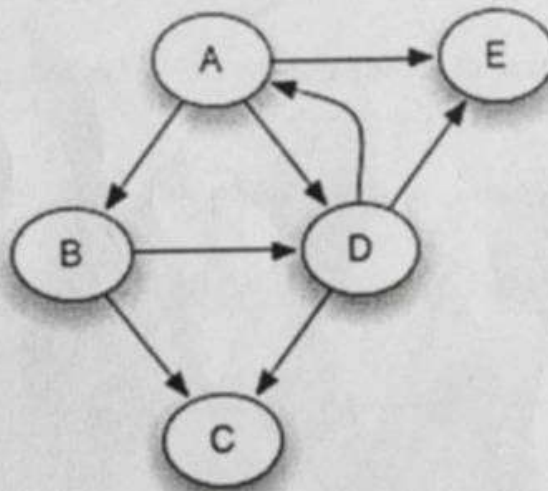


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Department of Computer Science and Engineering
Course Code: CSE 207 Title: Data Structures
Course Instructor: Manowarul Islam
Date: 17.05.2023 Spring 2023
Final Examination Section: 5
Time: 1 hour 20 minutes Marks:20

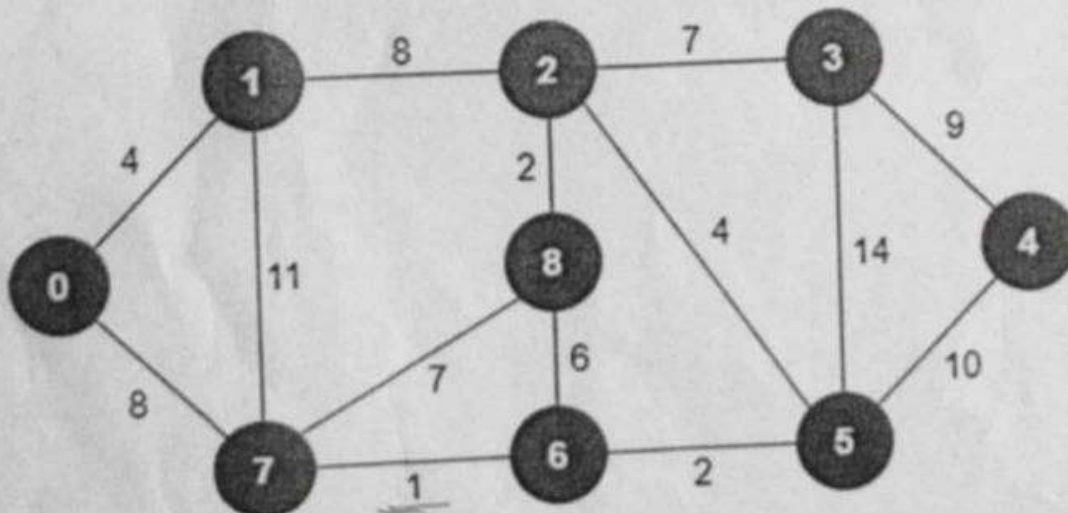
Note: Answer all questions. Figure in the right margin indicate marks.

1. Give an adjacency-list and adjacency-matrix representation of the following graph. Mention some application area of graphs. [CO2]-4

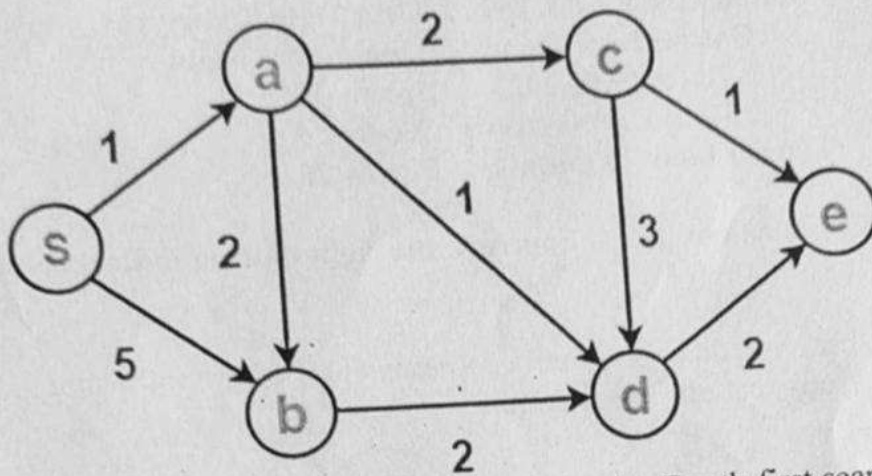


2. How you can detect a cycle in a graph. Explain with example. [CO2]-2.5

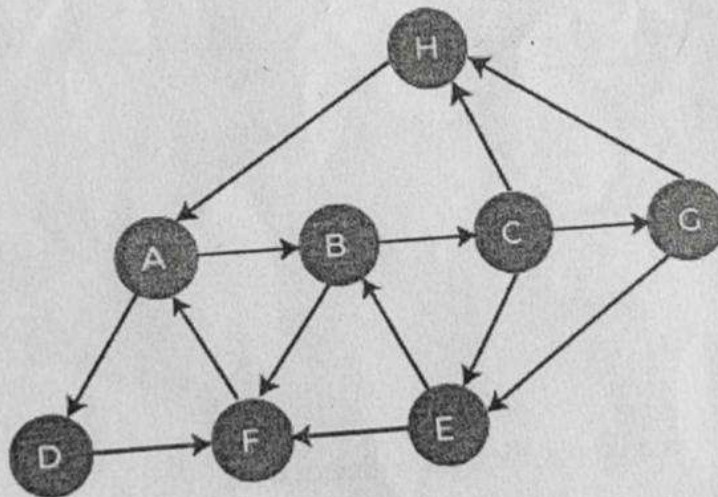
3. Find the MST using prim's algorithm. [CO3]-3.5



4. Find the single source shortest path using Dijkstra Algorithm. Assume source is S. [CO3]-5

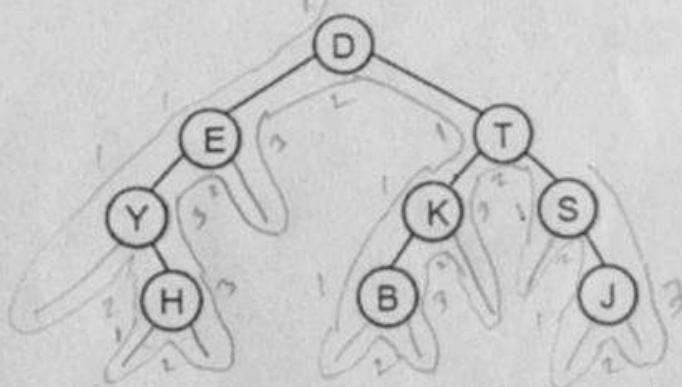


5. Show the d and f values that result from running the Depth-first search (DFS) on the directed graph given below. [CO2]-5



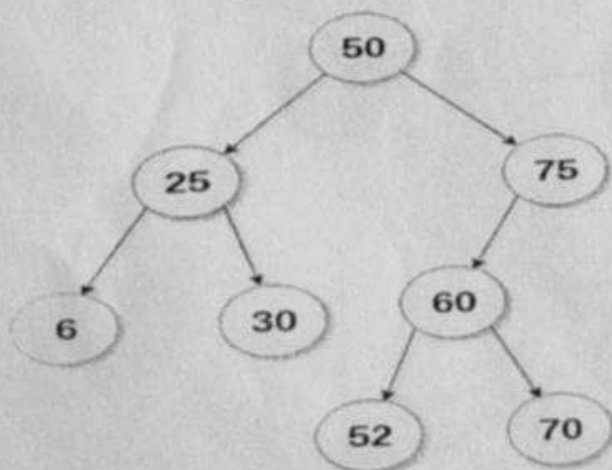
Note: Answer all questions. Figure in the right margin indicate marks.

1. What are the conditions of a well define recursion algorithm. 2
2. State the solution tree for tower of Hanoi problem using N=4. 3
3. What is BST? Draw a binary search tree using the following numbers: 4
 37, 2, 45, 48, 41, 29, 20, 30, 49, 70
4. Consider the following tree. 4
 i. Shows the sequential memory representation of the tree.
 ii. Show the in-order and post-order traversal of the tree.



Pre: Y, H, E, D, B, K, T, S, J
 Post: H, Y, E, B, K, J, S, T, D

5. Delete 50 from the following tree. Show the link representation before and after the delete operation. 5



6. Define extended binary tree with example. 2

Complete tree

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Department of Computer Science and Engineering

Course Code: CSE 207 Title: Data Structures

Course Instructor: Manowarul Islam

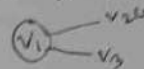
Date: 12.01.2023 Fall 2022

Final Examination Section: 5

Time: 1 hour 20 minutes Marks: 20

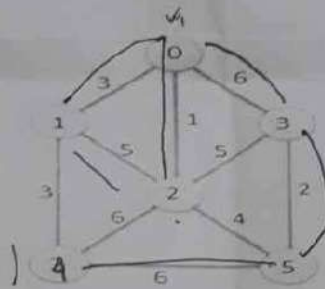
Note: Answer all questions. Figure in the right margin indicate marks.

- ✓ 1. Define connected graph with example. Consider the following graph $G = (V, E)$: $V = \{v_1, v_2, v_3, v_4, v_5, v_6, v_7\}$ and $E = \{v_1v_3, v_1v_5, v_2v_3, v_2v_5, v_3v_5, v_3v_6, v_4v_5, v_4v_6, v_5v_6, v_6v_7\}$. Draw the graph. [CO2]-4

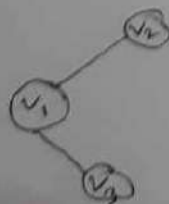
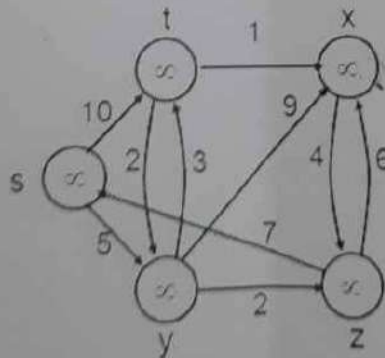


- ✓ 2. State recursive solution for the Tower of Hanoi problem if the number of Discs $N=4$. [CO2]-3

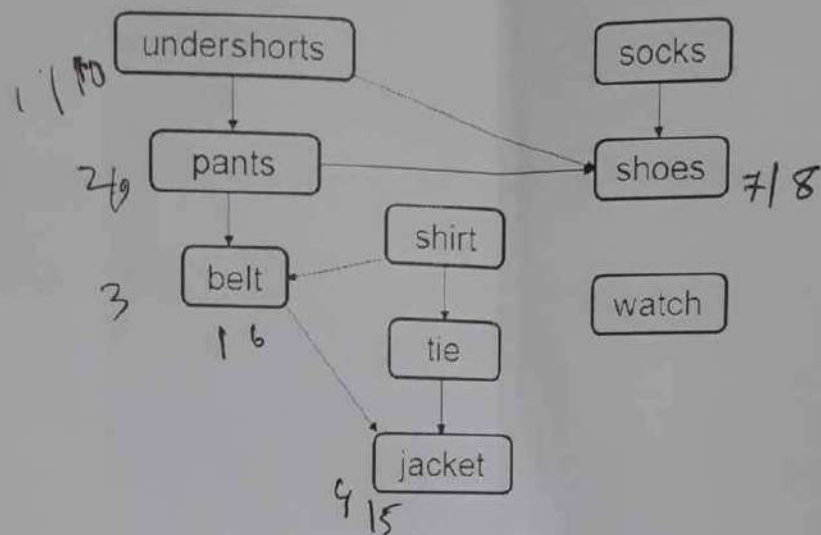
3. Find the MST using prim's algorithm. [CO3]-3



- ✓ 4. Find the single source shortest path using Dijkstra Algorithm. Assume source is S. [CO3]-4



- ✓ 5. Show the d and f values that result from running the Depth-first search (DFS) on the directed graph given below. [C]

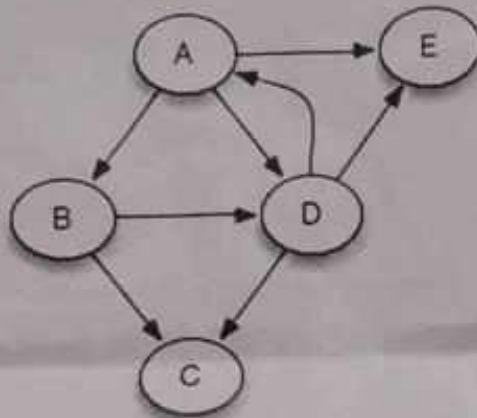


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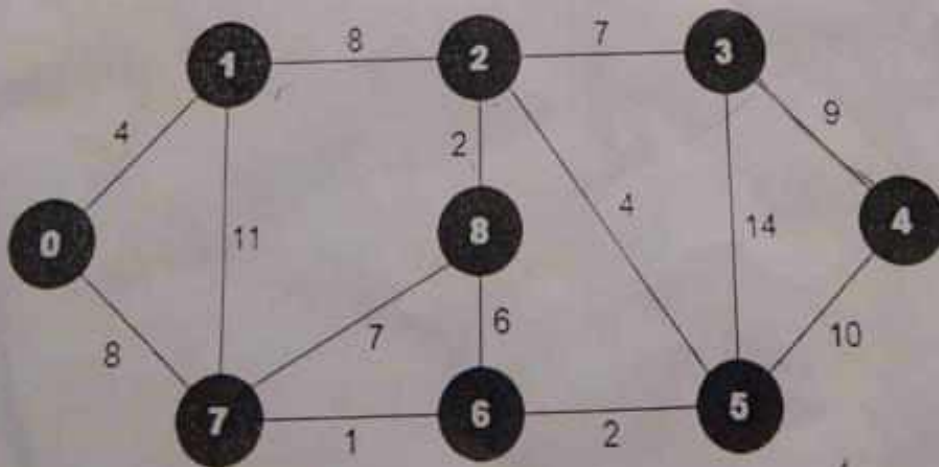
Department of Computer Science and Engineering
Course Code: CSE 207 Title: Data Structures
Course Instructor: Manowarul Islam
Date: 15.09.2022 Summer 2022
Final Examination Section: 5
Time: 1 hour 20 minutes Marks: 20

Note: Answer all questions. Figure in the right margin indicate marks.

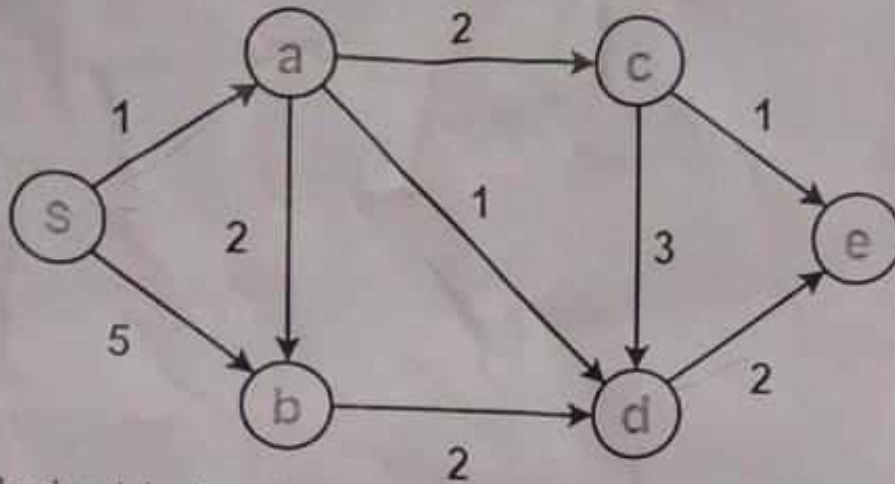
1. Give an adjacency-list and adjacency-matrix representation of the following [CO2]-2.5 graph.



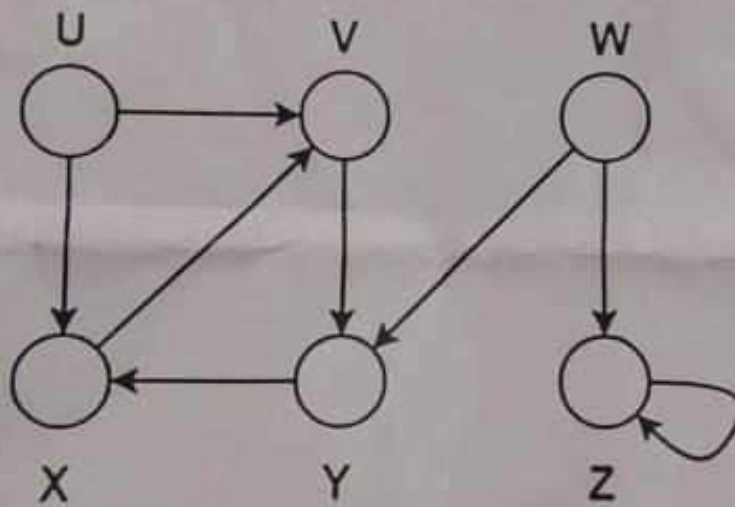
2. What is a Heap? Give example. [CO2]-2.5
3. Find the MST using prim's algorithm. [CO3]-3



4. Find the single source shortest path using Dijkstra Algorithm. Assume source is S. [CO3]-6



5. Show the d and f values that result from running the Depth-first search (DFS) on the directed graph given below (Show each steps clearly.). [CO2]-6

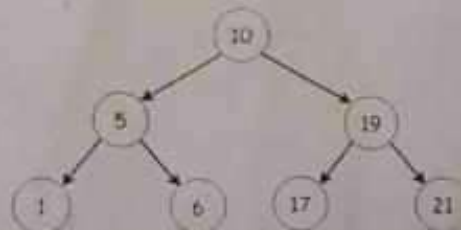


East West University

Department of Computer Science and Engineering
Course Code: CSE 207 Title: Data Structures
Course Instructor: Manowarul Islam
Date: 11.08.2022 Summer 2022
MID-2 Section: 5
Time: 1 hour 20 minutes Marks: 20

Note: Answer all questions. Figure in the right margin indicate marks.

1. Define Extended Binary Tree with example. [CO2]-3
2. What is BST? Draw a binary search tree using the following numbers: [CO2]-4
50, 70, 60, 20, 90, 10, 40, 100
3. Show the linked representation of the following tree: [CO2]-3

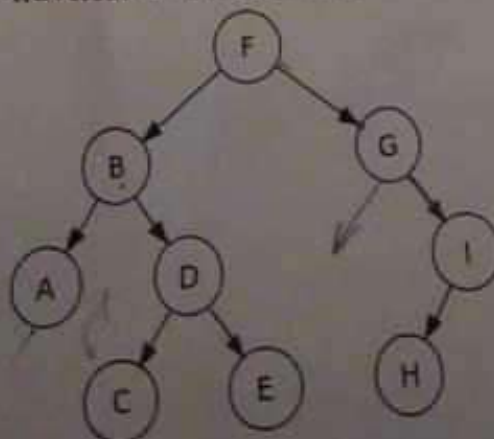


50740

4. How you can delete a node from BST that has one or two children. Explain with example. [CO3]-4
5. Find the Huffman tree of the following information. [CO3]-4

Letter	Freq
e	120
d	42
i	42
u	37
c	32
m	24
b	7
z	2

6. Show the post order traversal of the following tree. [CO3]-2

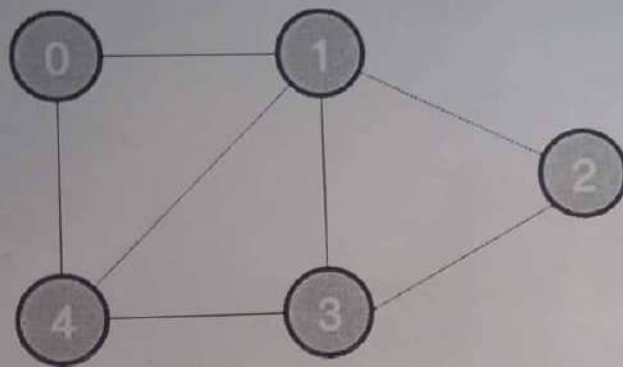


East West University

Department of Computer Science and Engineering
Course Code: CSE 207 Title: Data Structures
Course Instructor: Manowarul Islam
Date: 12.05.2022 Spring 2022
Final Examination Section: 5
Time: 1 hour 20 minutes Marks:20

Note: Answer all questions. Figure in the right margin indicate marks.

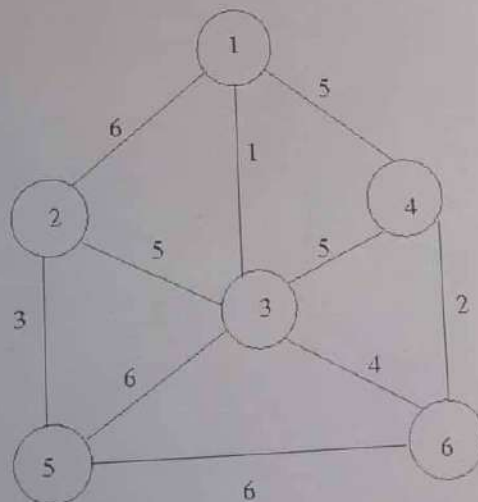
- ✓ 1. Give an adjacency-list and adjacency-matrix representation of the following [CO2]-2.5 graph.



2. What is a Heap? Give example.
3. Find the MST using prim's algorithm.

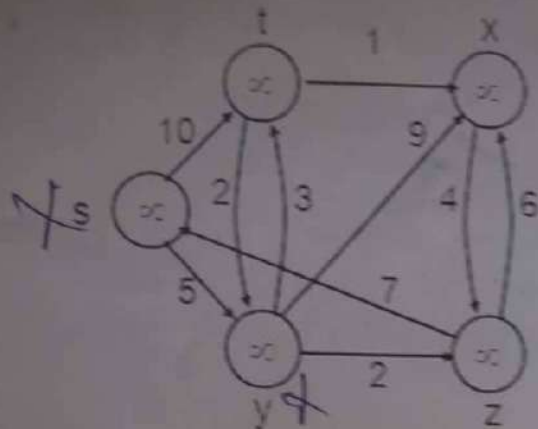
[CO2]-2.5

[CO3]-3

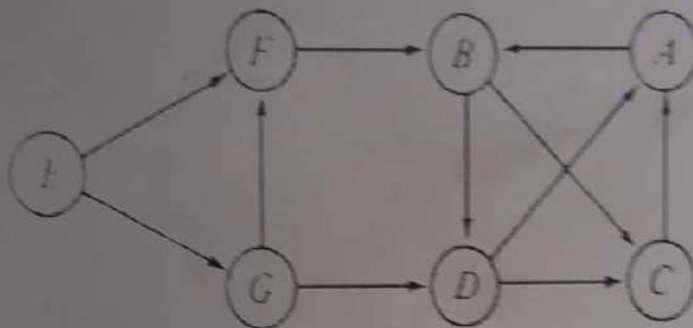


✓

4. Find the single source shortest path using Dijkstra Algorithm. Assume source is S. [CO3]-6

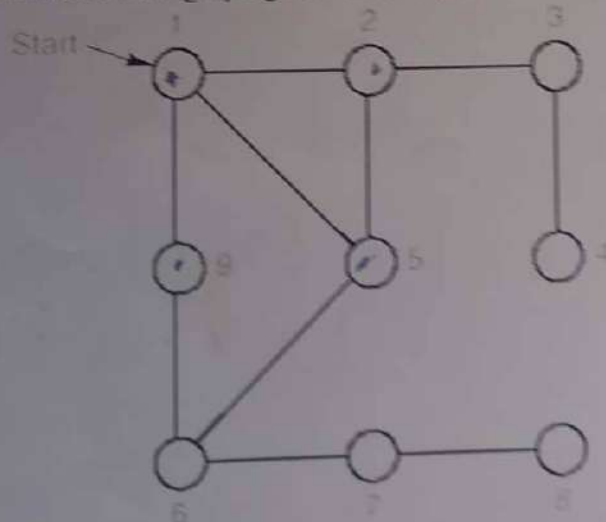


6. Show the d and f values that result from running the Depth-first search (DFS) on the directed graph given below. [CO2]-6



Or

Find the BFS forest for the graph given below: (Start from 1)

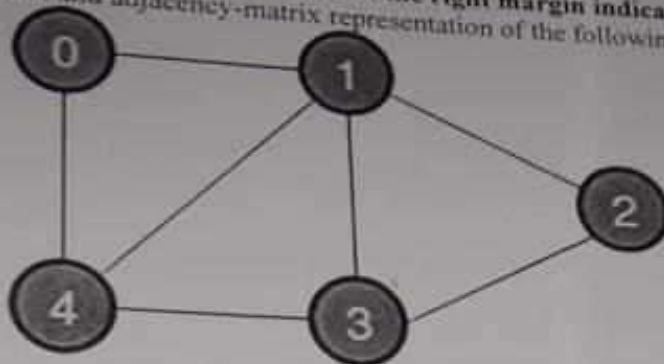


East West University

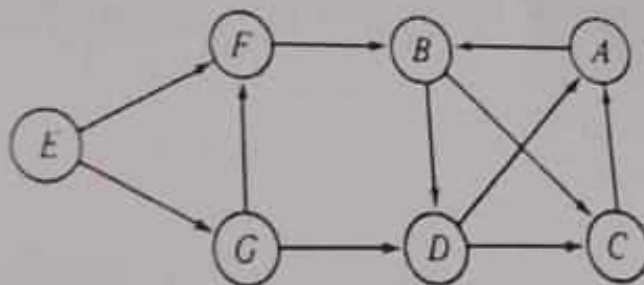
Department of Computer Science and Engineering
 Course Code: CSE 207 Title: Data Structures
 Course Instructor: Manowarul Islam
 CT 2

Note: Answer all questions. Figure in the right margin indicate marks.

1. Give an adjacency-list and adjacency-matrix representation of the following graph. 3

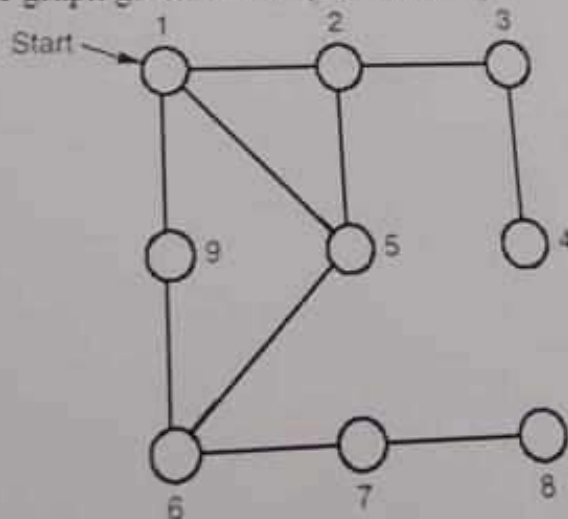


2. Show the d and f values that result from running the Depth-first search (DFS) on the directed graph given below. 7



Or

Find the BFS forest for the graph given below: (Start from 1)



East West University
Department of Computer Science and Engineering
Course Code: CSE 207 Title: Data Structures
Course Instructor: Manowarul Islam
Date: 03.03.2022 Fall 2022
ClassTest-1 Section: 5
Time: 20 minutes Marks:10

Note: Answer all questions. Figure in the right margin indicate marks.

- | | | |
|---|--|---|
| 1 | Define stack. Write down the basic steps for PUSH operation. | 3 |
| 2 | Convert the following expression to postfix
Q: $((A+B)*D) \uparrow (E-F)$. | 4 |
| 3 | Evaluate the following Postfix expression. P=45, 7, 2, -, /, 2, 1, 9, +, *, + | 3 |

East West University
Department of Computer Science and Engineering
Course Code: CSE 207 Title: Data Structures
Course Instructor: Manowarul Islam
Date: 03.03.2022 Fall 2022
Class Test-1 Section: 5
Time: 1 hour 20 minutes Marks: 20

Note: Answer all questions. Figure in the right margin indicate marks.

1. Find the postfix equivalent of the following expression and evaluate the result. **6**
 $(4+8) * (7-5) / ((4-2) * (2+2))$

2. Suppose you have a circular queue of size 6 (1 to 6). Initially the value of the front and rear is 0. Now show the state of the queue if the following operation is performed on the queue. Also find the value of front and rear in each step. **4**

insert (10); insert (5), delete (), insert (9); insert (9); insert (7); insert (5); insert (8); insert (10); insert (11).

3. State the memory representation of single linked list with example. **3**

4. Write a simple procedure to count the number of elements in a linked list. **4**

5. Linked lists have several advantages over arrays. Mention some advantages. **3**